# SPFresh: Incremental In-Place Update for Billion-Scale Vector Search

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## Abstract

Approximate Nearest Neighbor Search (ANNS) on high dimensional vector data is now widely used in various applications, including information retrieval, question answering, and recommendation. As the amount of vector data grows continuously, it becomes important to support updates to vector index, the enabling technique that allows for efficient and accurate ANNS on vectors.

Because of the curse of high dimensionality, it is often costly to identify the right neighbors of a new vector, a necessary process for index update. To amortize update costs, existing systems maintain a secondary index to accumulate updates, which are merged with the main index by globally rebuilding the entire index periodically. However, this approach has high fluctuations of search latency and accuracy, not to mention that it requires substantial resources and is extremely time-consuming to rebuild.

We introduce SPFresh, a system that supports in-place vector updates. At the heart of SPFresh is LIRE, a lightweight incremental rebalancing protocol to split vector partitions and reassign vectors in the nearby partitions to adapt to data distribution shifts. LIRE achieves low-overhead vector updates by only reassigning vectors at the boundary between partitions, where in a high-quality vector index the amount of such vectors is deemed small. With LIRE, SPFresh provides superior query latency and accuracy to solutions based on global rebuild, with only 1% of DRAM and less than 10% cores needed at the peak compared to the state-of-the-art, in a billion scale disk-based vector index with a 1% of daily vector update rate.

***CCS Concepts:*** • **Information systems → Information storage systems**; **Information retrieval**.

***Keywords:*** Vector Search, Incremental Update, Billion-scale



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## 1 Introduction

Today deep learning models can embed almost all types of data, including speech, vision, and text information, into *multi-dimensional vectors* with tens or even hundreds of dimensions. Such vectors are critical for complex semantic understanding tasks [42, 49]. To enable effective vector analysis, vector nearest neighbor search (NNS) systems have become critical system components for an increasing number of online services like search [35] and recommendation [57].

To satisfy the strict query latency requirement for these online services, vector search systems often resort to *approximate nearest neighbor search* (ANNS) [17, 22, 34, 51, 56, 59, 65, 69], to locate as many correct results as possible (i.e., query accuracy). At the heart of a large-scale NNS system is a *vector index*, a key data structure that organizes high-dimensional vectors efficiently for high-accuracy low-latency vector searches [3, 9, 14, 38, 58, 67].

Like traditional indices, a high-quality vector index organizes a quick "navigation map" of vectors based on the vector proximity in a high dimensional space. The proximity measurements are often implemented with "shortcuts", which only exist between a pair of vectors with a short distance. A search query traverses the datasets based on "shortcuts" the result set. The quality of the index for efficient traversal is highly dependent on the quality of shortcuts, where insufficient shortcuts miss relevant vectors, and extraneous shortcuts incur excessive traversal and storage costs. For high-dimensional data, vector indices require careful construction to produce a sufficient amount of high-quality "shortcuts", often as vector partitions [14, 67] or graphs of vector data [56, 65].

To add fuel to fire, there is a strong desire to support *fresh update* of vector indices because current systems generate a vast amount of vector data continuously in various settings. For example, 500+ hours of content are uploaded to

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